

Questions with Answer Keys

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Q1: 16 March (Shift 1) - Numerical



If the above equation is balanced with integer coefficients, the value of c is _____ (Round off to the Nearest Integer).

Q2: 16 March (Shift 2) - Single Correct

The exact volumes of 1M NaOH solution required to neutralise 50 mL of 1M H_3PO_3 solution and 100 mL of 2M H_3PO_2 solution, respectively, are:

- (1) 100 mL and 100 mL
- (2) 100 mL and 50 mL
- (3) 100 mL and 200 mL
- (4) 50 mL and 50 mL

Q3: 17 March (Shift 1) - Numerical

15 mL of aqueous solution of Fe^{2+} in acidic medium completely reacted with 20 mL of 0.03M aqueous $\text{Cr}_2\text{O}_7^{2-}$. The molarity of the Fe^{2+} solution is _____ $\times 10^{-2}\text{M}$ (Round off to the Nearest Integer).

Q4: 18 March (Shift 2) - Numerical

10.0ml of Na_2CO_3 solution is titrated against 0.2 M HCl solution. The following titre values were obtained in 5 readings.

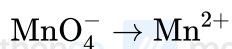
4.8ml, 4.9ml, 5.0ml, 5.0ml and 5.0ml Based on these readings, and convention of titrimetric estimation of concentration of Na_2CO_3 solution is _____ mM

(Round off to the Nearest integer)

Answer Key**Q1 (16)****Q2 (3)****Q3 (24)****Q4 (50)**

Q1 (16)

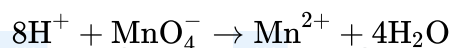
Writing the half reaction oxidation half reaction



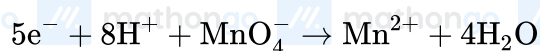
balancing oxygen



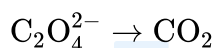
balancing Hydrogen



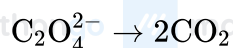
balancing charge



Reduction half



Balancing carbon



Balancing charge



Net equation

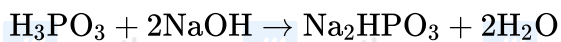
Hints and Solutions

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$$\text{So } c = 16$$

Q2 (3)

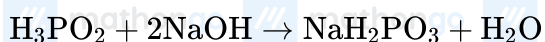


$$50\text{ml}^3 \quad 1\text{M}$$

$$1\text{M} \quad V = ?$$

$$\Rightarrow \frac{n_{\text{NaOH}}}{n_{\text{H}_3\text{PO}_3}} = \frac{2}{1}$$

$$\Rightarrow \frac{1 \times V}{50 \times 1} = \frac{2}{1} \Rightarrow V_{\text{NaOH}} = 100\text{ml}$$



$$100\text{ml}$$

$$2\text{M}$$

$$V = ?$$

$$\Rightarrow \frac{n_{\text{NaOH}}}{n_{\text{H}_3\text{PO}_2}} = \frac{1}{1} \Rightarrow \frac{1 \times V}{2 \times 100} = \frac{1}{1} \Rightarrow \boxed{V_{\text{NaOH}} = 200\text{ml}}$$

Q3 (24)

$$n_{\text{eq}}\text{Fe}^{2+} = n_{\text{eq}}\text{Cr}_2\text{O}_7^{2-}$$

$$\text{or, } \left(\frac{15 \times M_{\text{Fe}^{2+}}}{1000} \right) \times 1 = \left(\frac{20 \times 0.03}{1000} \right) \times 6$$

$$\therefore M_{\text{Fe}^{2+}} = 0.24\text{M} = 24 \times 10^{-2}\text{M}$$

Q4 (50)

Most precise volume of HCl = 5ml at equivalence point Meq. of Na_2CO_3 = meq. of HCl. Let molarity of

$$\text{Na}_2\text{CO}_3 \text{ solution} = M, \text{ then } M \times 10 \times 2 = 0.2 \times 5 \times 1$$

$$M = 0.05 \text{ mol/L}$$

$$= 0.05 \times 1000$$